**Supplementary Material**

# The Reach of Suction-Triggered Irrigation and the Marginal Fate of Irrigation Water: Closed Forms from the van Genuchten–Mualem Parameters

Derivations of the equations, the mechanisms behind the results, the data, and the origin of every table and figure

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## S1. Highlights

- A closed form from $\alpha$, $n$ and $\ell$ gives the largest suction a crop can impose at a sensor: the reach of a trigger.
- It predicts held-out soils within a factor of 2.3; triggers beyond it never fire and leave the crop rainfed.
- Of each irrigation millimetre, 38 % is transpired and 16 % drains; the rest evaporates.
- The drained share rises with the dose on a soil-independent scale of about 4 mm.
- Data (every log of every run), code, software and a 1155-page manual are open.

## S2. What is new in this article

The table sets each contribution against what was established before it. The derivations that follow mark which equations are taken from the literature and which are new.

**Table S1:** What this article adds, item by item: what was established before, what is new here, and the evidence in this article.

| # | Established before | New in this article | Evidence here |
|---|---|---|---|
| 1 | Field capacity has intrinsic definitions from the hydraulic parameters (Assouline and Or, 2014; Twarakavi et al., 2009); uptake becomes limited by the conductance of the soil around the roots (Carminati and Javaux, 2020; Gardner, 1960). | A closed form for the dry-end counterpart, the *reach* of a suction trigger, with one constant shared by all soils. | Held-out soils within a factor of 2.3 (LOSO 0.18 decades); a threshold on absolute conductivity rejected. |
| 2 | Triggers of 300 to 600 cm are tabulated for field crops (Allen et al., 1998). | These triggers lie beyond the reach of coarse soils: they never fire and the crop stays rainfed. | Sand and loamy sand at 300 cm: rainfed yield; a deeper sensor does not help (E3). |
| 3 | – | A reach-scaled trigger $h_t/h_c$ that places the irrigation responses of all soils on one axis. | Between-soil scatter falls from 54 % to 32 % of the total. |

| # | Established before | New in this article | Evidence here |
|---|---|---|---|
| 4 | Pulsing, discharge and hydraulic properties change losses below the root zone, shown case by case (Elmaloglou and Diamantopoulos, 2007, 2008, 2009). | The marginal fate of an irrigation millimetre across soils, doses, seasons and methods. | 0.38 transpired, 0.25 soil evaporation, 0.19 other evaporation, 0.16 drained. |
| 5 | – | A dose scale of the drained share that does not depend on the soil, and a soil-specific plateau. | $d_D = 4.1 \pm 0.6$ mm, $\rho = 0.07$ with $n$; plateau 0.10 to 0.45. |
| 6 | Interception losses of sprinklers are known. | Overhead irrigation adds no transpiration: intercepted water replaces it. | $\eta_T = -0.08$ (all soils) and $-0.01$ (coarse loam). |
| 7 | Flow can be out of equilibrium with the retention curve (Diamantopoulos and Durner, 2012; Ross and Smettem, 2000). | A criterion, $\Pi_\tau = \tau / t_i$, for when this changes the firing of a trigger (proposed, not tested). | Derivation in Section S3.8. |

## S3. The author's contributions: problem, derivation and pinpoints

Each contribution of the article is set out in the same way: *the problem* it answers, stated in a few lines; *the derivation*, step by step; *the result*, with the parts that matter marked by circled labels ①, ②, …; and *the pinpoints*, which say for each circled mark what it shows and why it is so. Numbers come from the data and can be recomputed in the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`.

### S3.1 Contribution 1: the reach of a suction trigger

**The problem**

A farmer on a sandy field installs a matric-potential sensor at 20 cm and sets the common trigger of 300 cm. All season the controller never opens the valve, the crop wilts, and the yield equals that of a field that was never irrigated. The sensor works. Why is the trigger never reached, and how dry a trigger can be used on any soil, from its van Genuchten parameters alone?

**The derivation**

**(a)** Roots dry a layer only as fast as the soil delivers water to them; the delivery is proportional to the conductivity $K = K_s K_r$ (Gardner, 1960). **(b)** Postulate: the sensor's soil stops drying when the relative conductivity has fallen to one value $\kappa$, the same for all soils, $K_r(h_c) = \kappa$. **(c)** At the dry end, $K_r \simeq m^2(\alpha h)^{-[2n+\ell(n-1)]}$ (Section S3.2). **(d)** Insert (c) into (b): $m^2(\alpha h_c)^{-E} = \kappa$ with $E = 2n + \ell(n-1)$. **(e)** Divide by $m^2$, raise to $-1/E$, divide by $\alpha$.

**The result**

$$h_c = \underbrace{\frac{1}{\alpha}}_{1} \left( \underbrace{m^2}_{2} \Big/ \underbrace{\kappa}_{3} \right)^{\overbrace{1/[\,2n+\ell(n-1)\,]}^{4}}$$

$$\underbrace{\kappa = 3.47 \times 10^{-10}}_{5}$$

$$\underbrace{\text{error on soils left out} \leq \text{a factor of } 2.3}_{6}$$

**Pinpoints: what each circled mark shows and why**

(1) $1/\alpha$ is the suction at which the soil starts to drain. It only *rescales* the reach: two soils with the same $n$ have reaches in the ratio of their $1/\alpha$. This is why, at equal $n$, a soil that starts draining later (small $\alpha$) can be dried further.

(2) $m^2 = (1 - 1/n)^2$ comes from the tail of the Mualem integral. It changes slowly with $n$ (0.035 to 0.39 over the ten soils) and enters under the root of order $E$, so its effect is small;

it is kept because it makes the formula exact at the dry end.

(3) $\kappa$ is the single constant of the law: the relative conductivity at which the crop can no longer dry the soil at the sensor within a season. It is a property of the crop and its uptake, not of the soil, which is why one value serves ten soils. It was fitted once and tested by leaving soils out.

(4) The exponent $1/E$ with $E = 2n + \ell(n-1)$ is the heart of the result. $E$ is the slope of $\log K_r$ against $\log h$ at the dry end. In sand ($n = 2.68$) $E = 6.2$: the conductivity collapses within a narrow band of suction, so the reach is short (199 cm). In silty clay loam ($n = 1.23$) $E = 2.6$: the conductivity fades slowly, so the reach is long ($1.3 \times 10^5$ cm). The pore-size distribution, through $n$, decides how far a crop can dry a soil.

(5) $\kappa = 3.47 \times 10^{-10}$ means the soil at the reach conducts about three billion times less than when saturated; with saturated conductivities of 1 to 700 cm d$^{-1}$ this is far below a millimetre per day, which is why uptake stops there.

(6) The law was tested on soils it had not seen: each soil was removed, $\kappa$ refitted on the other nine, and the removed soil predicted. Five predictions fell inside the observed interval and none was off by more than a factor of 2.3. A threshold on absolute conductivity, the obvious alternative, failed (it spans more than three decades).

### S3.2 Contribution 2: a rule for blind triggers

**The problem**

Which of the tabulated triggers can be used on which soil, before any season is lost?

**The derivation**

The irrigation of every soil fell to zero between $h_t/h_c = 0.5$ and 1 (Fig. 3b of the article). A trigger therefore fires reliably only below half the reach. Compute $h_c$ (Contribution 1), halve it, and compare with the trigger.

**The result**

$$\underbrace{h_t}_{(1)} < \underbrace{\tfrac{1}{2}}_{(2)} \underbrace{h_c(\alpha, n, \ell)}_{(3)}$$

| Soil | $h_c$ (cm) | $h_c/2$ (cm) | 300 cm trigger |
|---|---|---|---|
| Sand (S03) | 199 | 100 | (4) blind: never fires |
| Loamy sand (S04) | 426 | 213 | (4) blind: never fires |
| Sandy loam (S05) | 1 618 | 809 | fires |
| Loam (S06) | 9 209 | 4 605 | fires |
| Silty clay loam (S10) | 128 260 | 64 130 | fires |

**Pinpoints: what each circled mark shows and why**

(1) The trigger chosen by the user, in cm of suction at the sensor.

(2) The factor one half is a safety margin read from the data, not a fitted constant: below it every soil still irrigates.

(3) The reach from the soil's own parameters; no season of trial and error is needed.

(4) The two blind soils are exactly those in which the simulated 300 cm trigger never irrigated and the yield stayed at its rainfed value (1.13 and 1.38 Mg ha$^{-1}$, against 3.04 and 4.54 at 50 cm).

### S3.3 Contribution 3: one axis for all soils

**The problem**

Irrigation-against-trigger curves of ten soils lie far apart. Can one scale bring them together, so that one rule applies to all?

**The result**

$$\underbrace{x}_{(1)} = \frac{h_t}{h_c}, \qquad \text{scatter between soils: } \underbrace{54\,\%}_{(2)} \rightarrow \underbrace{32\,\%}_{(3)}$$

**Pinpoints: what each circled mark shows and why**

(1) The trigger measured in units of the soil's reach: $x = 1$ means "at the reach" in every soil.

(2) Share of the variance of the normalised irrigation that lies between soils when the raw trigger is used.

(3) The same share on the reach scale. The reach removes where irrigation *ends*; what remains is how fast it fades before the end, which depends on retention, not on conductivity.

### S3.4 Contribution 4: the fate of an irrigation millimetre

**The problem**

Of every millimetre of water pumped onto a field, how much does the crop transpire, and where does the rest go? The answer must hold for many soils, doses, seasons and methods, not for one case.

**The derivation**

Write the 20-year balance of an irrigated run and of its rainfed twin (same soil and weather), subtract them, and divide by the irrigation $I$ (Section S3.6).

**The result**

$$1 = \underbrace{\eta_T}_{(1)\ 0.38} + \underbrace{\eta_{E_s}}_{(2)\ 0.25} + \underbrace{\eta_{E_o}}_{(3)\ 0.19} + \underbrace{\eta_D}_{(4)\ 0.16} + \underbrace{\eta_S}_{(5)\ 0.03}, \qquad \eta_X = \frac{\sum(X - X_0)}{\sum I}$$

| Soil | transpired | soil evap. | other evap. | drained |
|---|---|---|---|---|
| Sandy loam | 0.64 | 0.11 | 0.14 | 0.07 |
| Loam | 0.41 | 0.22 | 0.17 | 0.15 |
| Coarse loam | 0.38 | 0.32 | 0.19 | 0.08 |
| Coarse sand | 0.23 | 0.19 | 0.26 | (6) 0.32 |

**Pinpoints: what each circled mark shows and why**

(1) Only 38 % of irrigation becomes transpiration, the only flux that makes yield. Transpiration is capped by the energy reaching the canopy and by its cover.

(2) A quarter evaporates from the wetted soil surface: water applied to bare or partly covered soil uses the same energy as transpiration.

(3) Another fifth evaporates from ponded water, litter and intercepted water, the term usually left out of irrigation accounts.

(4) Sixteen per cent drains below 1 m and is lost to the crop (and carries nitrate).

(5) The small remainder changes storage or runs off; it closes the balance and is a check of the calculation.

(6) In coarse sand drainage is the largest single loss: the soil passes extra water on before the crop can use it.

### S3.5 Contribution 5: the dose law

**The problem**

Small and frequent doses are recommended to reduce deep percolation. How small is small, and does the answer depend on the soil?

**The derivation**

For each soil the drained share was computed at ten doses (1 to 30 mm). A form was sought that is zero for a vanishing dose and levels off for a large one; the simplest is the saturating exponential, fitted per soil ($R^2$ 0.92 to 0.96 on seven soils).

**The result**

$$\eta_D(d) = \underbrace{\eta_\infty}_{(1)}\left(1 - e^{-d/\underbrace{d_D}_{(2)}}\right)$$

$$\underbrace{\eta_D \approx \eta_\infty\, d/d_D \;\; (d \ll d_D)}_{(3)}$$

$$\underbrace{d_D = 4.1 \pm 0.6 \text{ mm},\; \rho(d_D, n) = 0.07}_{(4)}$$

**Pinpoints: what each circled mark shows and why**

(1) The plateau: the share of a large dose that drains. It differs between soils (0.10 to 0.20, and 0.45 in coarse sand) because it is the outcome of a race between drainage and uptake,

which the soil's conductivity decides.

(2) The dose scale: the dose at which 63 % of the plateau is reached. It is about 4 mm in every soil.

(3) For small doses the drained share grows in proportion to the dose: a dose much smaller than $d_D$ hardly drains, but in exchange most of it evaporates (60 to 72 % at 1 mm).

(4) $d_D$ is not related to $n$ or to any soil property: it is set by the atmosphere and the crop. Four millimetres is about one day of crop water use, so a dose below one day's use is consumed before it can move below the roots.

### S3.6 Contribution 6: overhead irrigation adds no transpiration

**The result**

$$\underbrace{\eta_T^{\text{overhead}}}_{(1)} = -0.08 \text{ (all soils)}$$

$$\underbrace{\eta_{E_o}^{\text{overhead}}}_{(2)} = 0.79$$

**Pinpoints: what each circled mark shows and why**

(1) A negative marginal transpiration: the crop transpires slightly *less* with overhead irrigation than without.

(2) Most of the water is intercepted by the canopy and evaporates with the energy that would have driven transpiration; interception replaces transpiration instead of adding to it. Subsurface application avoids this (0.51 transpired) but drains more (0.23).

### S3.7 Contribution 7: when a sensor's lag matters

**The result**

$$\underbrace{\Pi_\tau}_{(1)} = \frac{\overbrace{\tau}^{(2)}}{\underbrace{t_i}_{(3)}}, \qquad \delta h \simeq \underbrace{\frac{r\tau}{C(h)}}_{(4)}$$

**Pinpoints: what each circled mark shows and why**

(1) A dimensionless number: if it is much smaller than one, the equilibrium reach of Contribution 1 holds.

(2) The relaxation time of the water content towards the retention curve (minutes to hours).

(3) The interval between irrigations. Daily scheduling gives $\Pi_\tau \approx 0.02$; pulses every half hour give $\Pi_\tau \approx 2$.

(4) The water capacity $C = -\mathrm{d}\theta/\mathrm{d}h$. It is smallest on the steep dry limb of a sand, so the same lag gives the largest error exactly where the reach is shortest. This criterion is proposed, not tested: Daisy solves equilibrium flow.

## S4. A step-by-step guide for students

This section explains, with numbers that can be checked on a calculator or in the workbook (sheets `EQ_P4_reach`, `EQ_P4_fate`, `EQ_P4_dose`), how every result of the article is obtained. Only school algebra, logarithms and the exponential function are needed. The full mathematical derivations follow in the next section.

### S4.1 Step 1: what a retention curve says

A soil holds water in pores of many sizes. When the soil dries, the large pores empty first because water is held in them by weaker capillary forces. The *suction* $h$ (in cm of water) measures how strongly the remaining water is held: 0 cm means saturated, 100 cm is about field capacity, 15 000 cm is about the wilting point. The *effective saturation* $S_e$ is the fraction of the movable water still in the soil (1 = full, 0 = empty). The van Genuchten curve links the two:

$$S_e = \left[1 + (\alpha h)^n\right]^{-m}, \qquad m = 1 - 1/n.$$

$\alpha$ (1/cm) tells at which suction the soil starts to drain, $n$ tells how sharply it drains: a sand has a large $n$ (it drains almost all at once), a clay a small $n$ (it drains slowly over a wide range). *Example (sand, $\alpha = 0.145$, $n = 2.68$).* $m = 1 - 1/2.68 = 0.627$. At $h = 100$ cm: $\alpha h = 14.5$; $14.5^{2.68} = 1\,293$; $1 + 1\,293 = 1\,294$; $1\,294^{-0.627} = 0.011$. The sand at field capacity holds only 1.1 % of its movable water.

### S4.2 Step 2: what the conductivity curve says

Water moves through filled pores. As the soil dries, fewer and thinner pores carry water, so the *relative conductivity* $K_r$ (1 = saturated) falls very fast. Mualem's formula gives

$$K_r = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2.$$

*Example (sand at 100 cm).* $S_e = 0.0112$, $S_e^{1/m} = 0.0112^{1.595} = 7.7 \times 10^{-4}$; $1 - 7.7 \times 10^{-4} = 0.99923$; $0.99923^{0.627} = 0.99952$; $1 - 0.99952 = 4.8 \times 10^{-4}$; squared $= 2.3 \times 10^{-7}$; times $\sqrt{0.0112} = 0.106$ gives $K_r \approx 2.5 \times 10^{-8}$. At field capacity the sand already conducts 40 million times less than when saturated.

### S4.3 Step 3: why a crop cannot dry a soil beyond a certain suction

Roots can only take water as fast as the soil brings it to them, and the soil brings water in proportion to its conductivity. Once $K_r$ falls below a tiny value $\kappa$, the soil around the roots stops delivering water, the crop is stressed, and the soil at the sensor does not dry any further. The suction where this happens is the *reach* $h_{\mathrm{c}}$ of the soil. A trigger set drier than the reach is never read by the sensor, so the field is never irrigated.

### S4.4 Step 4: a simple formula for the conductivity of a dry soil

When the soil is dry ($\alpha h$ much larger than 1), the 1 in the retention curve can be neglected, and the conductivity becomes a straight line on a log–log plot:

$$K_r \approx m^2(\alpha h)^{-E}, \qquad E = 2n + \tfrac{1}{2}(n-1).$$

*Check (sand).* $E = 2(2.68) + 0.5(1.68) = 6.2$; $m^2 = 0.393$; at $h = 100$ cm, $14.5^{-6.2} = 6.3 \times 10^{-8}$; $0.393 \times 6.3 \times 10^{-8} = 2.5 \times 10^{-8}$, the same as Step 2.

### S4.5 Step 5: solve for the reach

Set $K_r(h_{\mathrm{c}}) = \kappa$ in Step 4 and solve for $h_{\mathrm{c}}$: take both sides to the power $-1/E$, then divide by $\alpha$:

$$h_{\mathrm{c}} = \frac{1}{\alpha}\Big(\frac{m^2}{\kappa}\Big)^{1/E}.$$

The constant $\kappa$ is found once from the simulations (Step 6) and is the same for every soil: $\kappa = 3.47 \times 10^{-10}$. *Example (sand).* $m^2/\kappa = 0.393/3.47 \times 10^{-10} = 1.13 \times 10^9$. Its natural logarithm is 20.85; divided by $E = 6.2$ gives 3.363; $e^{3.363} = 28.9$; divided by $\alpha = 0.145$ gives $h_c = 199$ cm. *Example (loam, $\alpha = 0.036$, $n = 1.56$).* $m = 0.359$, $m^2 = 0.129$, $E = 3.4$; $\ln(0.129/3.47 \times 10^{-10}) = 19.73$; $19.73/3.4 = 5.80$; $e^{5.80} = 331$; $331/0.036 = 9\,209$ cm. So the sand's reach is 199 cm and the loam's is 9 209 cm: a 300 cm trigger works on loam but is never reached on sand.

### S4.6 Step 6: how $\kappa$ was found and tested

For each soil the simulations give an interval for the reach: the driest trigger that still irrigates (at least 5 mm a year) and the next drier trigger, which does not. For sand that is 100 to 150 cm. A value of $\kappa$ is tried, the reach of every soil is computed with Step 5, and the distance (in powers of ten) between each prediction and its interval is squared and added; $\kappa$ is changed until this sum is smallest. To test the formula fairly, each soil is left out in turn, $\kappa$ is found from the other nine, and the left-out soil is predicted. The worst prediction was a factor of 2.3 from its interval. In the workbook, change $\kappa$ in cell B2 of `EQ_P4_reach` and watch every reach and error change.

### S4.7 Step 7: where an irrigation millimetre goes

Compare an irrigated run with the same soil and weather without irrigation (the rainfed baseline). The differences in each water flux, divided by the irrigation, tell what each millimetre became. *Example: one run on coarse loam* (dose 10 mm, April to October). Irrigation $I = 222.5$ mm a year. Transpiration rose from 147.6 to 223.0 mm, $+75.4$ mm, so $\eta_T = 75.4/222.5 = 0.34$. Drainage rose from 6.1 to 39.5 mm, so $\eta_D = 33.4/222.5 = 0.15$. Soil evaporation rose from 146.4 to 228.8 mm, so $\eta_{E_s} = 0.37$. Other evaporation (ET minus transpiration minus soil evaporation) rose from 172.8 to 193.3 mm, so $\eta_{E_o} = 0.09$. The four add to 0.95; the remaining 0.05 went to storage and runoff. *Pooling the 100 irrigated runs of that soil.* Add all transpiration (21 328 mm), subtract 100 times the rainfed value ($100 \times 147.6 = 14\,762$), and divide by all irrigation (17 277 mm): $\eta_T = 6\,567/17\,277 = 0.38$. The sheet `EQ_P4_fate` does exactly this with `SUMIFS` for every soil.

### S4.8 Step 8: the dose law

A small dose is used by the crop and the surface on the same day and cannot reach the deep soil; a large dose can. The share that drains therefore starts near zero and grows towards a ceiling $\eta_\infty$:

$$\eta_D(d) = \eta_\infty\big(1 - e^{-d/d_D}\big).$$

*Example (loam, $\eta_\infty = 0.197$, $d_D = 3.9$ mm).* For a 1 mm dose: $1/3.9 = 0.26$, $e^{-0.26} = 0.77$, $1 - 0.77 = 0.23$, times 0.197 gives 0.045. For 10 mm: $e^{-2.6} = 0.074$, so $0.197 \times 0.926 = 0.18$. For 30 mm: almost the ceiling, 0.197. The dose scale $d_D$ was about 4 mm on every soil, close to one day of crop water use; the ceiling differs between soils (0.10 to 0.45).

### S4.9 Step 9: when the sensor's lag matters

In fast wetting and drying the water content can lag behind the value of the retention curve by a time $\tau$. If the schedule irrigates every $t_i$ days, the lag matters only when $\Pi_\tau = \tau/t_i$ is not small. *Example:* $\tau = 1$ h and irrigation every 2 days gives $\Pi_\tau = 1/48 = 0.02$, negligible; pulses every 30 min give $\Pi_\tau = 2$, important.

### S4.10 Step 10: reproduce everything

Open `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`: the yellow cells are the inputs, the green cells the formulas of Steps 1 to 8. Or run `Manuscripts/reproducibility/analysis/p4_reach.py`, which reads every run and writes every number and figure of the article.

## S5. Derivations of the equations

Sections S3.1 and S3.2 re-derive the van Genuchten–Mualem pair and its dry-end asymptote, which are established mathematics (Mualem, 1976; van Genuchten, 1980). Everything from S3.3 on is new in this article: the reach of a trigger and its properties, the alternative laws and their test, the interval-censored fit, the marginal-fate identity applied across soils, doses, seasons and methods, the dose law, and the non-equilibrium criterion for triggers (the relaxation equation itself is established (Ross and Smettem, 2000)).

### S5.1 The van Genuchten–Mualem pair from a capillary bundle

A soil is idealised as a bundle of capillaries whose radii are distributed so that the water-filled fraction at suction $h$ is the effective saturation $S_e$. The capillary (Young–Laplace) law relates the radius $r$ of the largest filled pore to the suction, $r = 2\sigma\cos\gamma/(\rho_w g h)$, so $h \propto 1/r$. Mualem (1976) wrote the relative conductivity of such a medium as the product of a tortuosity–connectivity term $S_e^{\ell}$ and the square of a normalised integral of the pore radius over the filled pores:

$$K_r(S_e) = S_e^{\ell}\left[\frac{\int_0^{S_e} h(x)^{-1}\,\mathrm{d}x}{\int_0^{1} h(x)^{-1}\,\mathrm{d}x}\right]^2 . \tag{S1}$$

van Genuchten (1980) chose the retention function $S_e = [1 + (\alpha h)^n]^{-m}$, whose inverse is

$$h(x) = \frac{1}{\alpha}\left(x^{-1/m} - 1\right)^{1/n}. \tag{S2}$$

Substituting Eq. (S2) and $y = x^{1/m}$ ($x = y^m$, $\mathrm{d}x = m\,y^{m-1}\mathrm{d}y$):

$$\int_0^{S_e} \frac{\mathrm{d}x}{h(x)} = \alpha m \int_0^{S_e^{1/m}} \left(y^{-1} - 1\right)^{-1/n} y^{m-1}\,\mathrm{d}y = \alpha m \int_0^{S_e^{1/m}} y^{m-1+1/n}(1-y)^{-1/n}\,\mathrm{d}y. \tag{S3}$$

With the restriction $m = 1 - 1/n$ the exponent of $y$ vanishes and $(1-y)^{-1/n} = (1-y)^{m-1}$, so the integral is elementary:

$$\int_0^{S_e} \frac{\mathrm{d}x}{h(x)} = \alpha m \int_0^{S_e^{1/m}} (1-y)^{m-1}\mathrm{d}y = \alpha\left[1 - \left(1 - S_e^{1/m}\right)^m\right]. \tag{S4}$$

The denominator of Eq. (S1) is the same expression at $S_e = 1$, that is $\alpha$, and the closed form of the paper follows:

$$K_r(S_e) = S_e^{\ell}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2. \tag{S5}$$

This is why the restriction $m = 1 - 1/n$ is not a convenience: it is the condition under which the Mualem integral has a closed form.

### S5.2 The dry-end asymptote of the conductivity

For $\alpha h \gg 1$, $(\alpha h)^n$ dominates the bracket of the retention function and

$$S_e = \left[1 + (\alpha h)^n\right]^{-m} = (\alpha h)^{-mn}\left[1 + (\alpha h)^{-n}\right]^{-m} = (\alpha h)^{-(n-1)}\Big[1 - m(\alpha h)^{-n} + O\big((\alpha h)^{-2n}\big)\Big], \tag{S6}$$

since $mn = n - 1$. Write $u = S_e^{1/m}$, which is small. The binomial expansion $(1-u)^m = 1 - mu - \frac{m(1-m)}{2}u^2 + O(u^3)$ gives

$$1 - (1-u)^m = mu\Big[1 + \tfrac{1-m}{2}u + O(u^2)\Big]. \tag{S7}$$

Inserting into Eq. (S5),

$$K_r = m^2\, S_e^{\ell+2/m}\Big[1 + (1-m)S_e^{1/m} + \dots\Big]. \tag{S8}$$

The exponent is $(\ell + 2/m)(n-1) = \ell(n-1) + 2n$, because $(n-1)/m = n$. Hence

$$K_r(h) = m^2(\alpha h)^{-[2n+\ell(n-1)]}\Big[1 + \varepsilon(h)\Big], \qquad \varepsilon(h) = O\big((\alpha h)^{-n}\big) + O\big((\alpha h)^{-n}\big), \tag{S9}$$

where both correction terms are of the order $(\alpha h)^{-n}$ (the first from the retention bracket, the second from $S_e^{1/m} = (\alpha h)^{-n}[1 + \dots]$). For the soils of the paper the reach lies at $\alpha h_c$ between 29 and 1 280, where $(\alpha h_c)^{-n}$ is below $10^{-3}$: the asymptote is exact to better than one per mille at the reach.

### S5.3 The reach of a trigger

*The physical condition.* During a drying cycle the crop takes water from the layer that holds the sensor at a rate $u$ limited by the conductance of the soil between the bulk soil and the root surface (Gardner, 1960). For a cylindrical root of radius $r_0$ in a soil cylinder of radius $r_1$, steady radial flow gives

$$u \simeq \frac{2\pi L_v}{\ln(r_1/r_0)} \int_{h_{\text{root}}}^{h} K(h')\,\mathrm{d}h' \;\approx\; \frac{2\pi L_v}{\ln(r_1/r_0)}\, K(h)\,\Delta h, \tag{S10}$$

with $L_v$ the root length density and $\Delta h$ the head drop to the root. Because $K$ falls by many decades while $\Delta h$ is bounded by the plant (the wilting point), $u$ collapses once $K(h)$ is small enough, and the soil at the sensor stops drying within the time of a season. The suction where

this happens is the reach $h_c$.

*One constant for all soils.* If the reach is reached at one value $\kappa$ of the relative conductivity, $K_r(h_c) = \kappa$, Eq. (S9) gives

$$m^2(\alpha h_c)^{-[2n+\ell(n-1)]} = \kappa \quad \Longrightarrow \quad \boxed{h_c = \frac{1}{\alpha}\left(\frac{m^2}{\kappa}\right)^{1/[2n+\ell(n-1)]}} \tag{S11}$$

*Properties.* (i) $\alpha$ only rescales the reach: $\alpha h_c$ depends on $n$, $\ell$ and $\kappa$ alone. (ii) The logarithmic sensitivity to $n$ is

$$\frac{\partial \ln(\alpha h_c)}{\partial n} = \frac{2}{m\, n^2\, [2n + \ell(n-1)]} - \frac{(2+\ell)\ln(m^2/\kappa)}{[2n+\ell(n-1)]^2}, \tag{S12}$$

whose second term dominates (with $\kappa = 3.5 \times 10^{-10}$, $\ln(m^2/\kappa) \approx 18$ to 21), so the reach shortens steeply as $n$ grows: from $\alpha h_c \approx 1\,280$ at $n = 1.23$ to 29 at $n = 2.68$. (iii) A larger $\ell$ shortens the reach because the conductivity collapses faster. (iv) For $n \to 1$ the exponent tends to 2 and $m \to 0$, so the reach tends to $\alpha^{-1}(m^2/\kappa)^{1/2}$, which vanishes slowly: very broad pore-size distributions are still reachable far into the dry range, as observed for the silty clay loam.

### S5.4 The alternative laws

*Fixed saturation.* If the reach were set by a water content, $S_e(h_c) = S_c$, inverting the retention function gives $h_c = \alpha^{-1}(S_c^{-1/m} - 1)^{1/n}$. It ignores the conductivity and predicts the coarse soils worst (Table 2 of the paper).

*Power law in $n-1$.* $\alpha h_c = A(n-1)^{-B}$ is an empirical two-parameter law; it fits slightly better but has no derivation.

*Absolute conductivity.* $K(h_c) = K_s K_r(h_c) = q_c$ would give $h_c = \alpha^{-1}(m^2 K_s/q_c)^{1/[2n+\ell(n-1)]}$. With literature $K_s$ the conductivity at the observed reaches spans more than three decades, so a single $q_c$ is rejected.

### S5.5 Fitting interval-censored reaches

For each soil the data give an interval $[\,h_{\text{lo}}, h_{\text{hi}})$: $h_{\text{lo}}$ is the driest trigger that still brings at least 5 mm of irrigation a year and $h_{\text{hi}}$ the next drier trigger, or $+\infty$ when even the driest trigger fires.

The loss of a prediction $\hat{h}$ is the squared distance in decades to the interval,

$$\mathcal{L}(\hat{h}) = \begin{cases} (\log_{10} h_{\mathrm{lo}} - \log_{10} \hat{h})^2, & \hat{h} < h_{\mathrm{lo}}, \\ 0, & h_{\mathrm{lo}} \le \hat{h} \le h_{\mathrm{hi}}, \\ (\log_{10} \hat{h} - \log_{10} h_{\mathrm{hi}})^2, & \hat{h} > h_{\mathrm{hi}}, \end{cases} \tag{S13}$$

summed over soils and minimised over $\log_{10} \kappa$ (Nelder–Mead). Leave-one-soil-out validation refits $\kappa$ without the soil and evaluates $\mathcal{L}$ on it; the reported error is $\sqrt{\mathcal{L}}$ in decades and the worst factor is $10^{\max \sqrt{\mathcal{L}}}$.

### S5.6 The marginal fate of irrigation

For a run with irrigation $I$ and its rainfed twin (same soil, weather and crop), the 20-year water balances are

$$P + I = T + E_s + E_o + D + \Delta S + Q, \qquad P = T_0 + E_{s,0} + E_{o,0} + D_0 + \Delta S_0 + Q_0. \tag{S14}$$

Subtracting and dividing by $I$ gives the identity of the paper,

$$1 = \underbrace{\frac{T - T_0}{I}}_{\eta_T} + \underbrace{\frac{E_s - E_{s,0}}{I}}_{\eta_{E_s}} + \underbrace{\frac{E_o - E_{o,0}}{I}}_{\eta_{E_o}} + \underbrace{\frac{D - D_0}{I}}_{\eta_D} + \underbrace{\frac{\Delta S - \Delta S_0 + Q - Q_0}{I}}_{\eta_S}. \tag{S15}$$

The shares are marginal, not average: a share can be negative (overhead irrigation lowers transpiration) or exceed the average loss fraction. Pooling a group of runs uses the ratio of sums, $\eta_X = \sum (X - X_0) / \sum I$, which weights each run by its irrigation and is therefore the share of the group's total irrigation.

### S5.7 The dose law

Let a dose $d$ be applied when the root zone has a deficit. Part of the dose is used by the crop and the surface before it can move below the roots; the rest can drain. If the water that can be used before redistribution grows with the dose and saturates, and the drained share of the remainder is a soil property $\eta_\infty$, the simplest form with the right limits is

$$\eta_D(d) = \eta_\infty \big(1 - e^{-d/d_D}\big), \qquad \eta_D \to \eta_\infty \frac{d}{d_D} \ (d \ll d_D), \qquad \eta_D \to \eta_\infty \ (d \gg d_D). \tag{S16}$$

The small-dose limit is linear because a vanishing dose cannot drain; the large-dose limit is the share of any extra water that the soil passes on. The dose scale $d_D$ is the dose at which $1 - e^{-1} = 63\,\%$ of the plateau is reached. The form is phenomenological: it was chosen for its limits and tested against the data (coefficient of determination 0.92 to 0.96 on seven soils), not derived from the Richards equation.

### S5.8 Dynamic non-equilibrium and the lag of a sensor

In the effective approach of Ross and Smettem (2000) and Diamantopoulos et al. (2012) the water content relaxes towards the equilibrium content of the retention curve with a time constant $\tau$:

$$\frac{\partial \theta}{\partial t} = \frac{\theta_{\mathrm{eq}}(h) - \theta}{\tau}. \tag{S17}$$

During a drying phase in which $\theta_{\mathrm{eq}}$ falls at a steady rate $r = -\mathrm{d}\theta_{\mathrm{eq}}/\mathrm{d}t$, the solution after the transient is $\theta = \theta_{\mathrm{eq}} + r\tau$, so the soil is wetter than equilibrium by $r\tau$, and a sensor that infers suction from the equilibrium curve under-reads the suction by

$$\delta h \simeq \frac{r\tau}{C(h)}, \qquad C(h) = -\frac{\mathrm{d}\theta_{\mathrm{eq}}}{\mathrm{d}h}. \tag{S18}$$

Over a schedule interval $t_i$ the drying is $rt_i$, so the relative error in the time of firing is of order $\Pi_\tau = \tau/t_i$, the criterion of the paper. Where $C(h)$ is small (the steep dry limb of a sand), $\delta h$ is large for the same $r\tau$, so non-equilibrium matters most where the reach is shortest.

## S6. Why the results happen

*Why a trigger has a reach.* The crop can only dry the soil at the sensor as fast as the soil delivers water to the roots. That delivery scales with the conductivity, which falls by many orders of magnitude as the soil dries. In a soil with a narrow pore-size distribution (large $n$, sands) the conductivity collapses within a narrow band of suction just past the air entry, so the crop stops drying the soil there; whatever trigger is set beyond that band is never read. In a soil with a broad distribution (small $n$, silty clay loams) the conductivity fades slowly over many decades of suction and the crop can still dry the soil at very high suctions.

*Why a deeper sensor does not help.* Below the sensor depth of the experiments the root density is smaller, so the soil dries even less, and in sands the deeper soil is also wetted by drainage from above. A deeper sensor therefore reads lower suctions, not higher ones.

*Why scaling by the reach collapses the curves only in part.* The reach sets where irrigation ends, because it is fixed by the conductivity. How irrigation fades before the reach depends on how often the soil reaches a given suction in a season, which depends on the water held between field capacity and that suction, a retention property that differs from the conductivity property that sets the reach.

*Why less than half of an irrigation millimetre is transpired.* Transpiration is limited by the energy available to the canopy and by the canopy cover. Water applied to a wet surface or to a canopy that does not cover the ground evaporates from the soil, from the ponded surface and from intercepted water, using the same energy; water applied beyond the storage of the root zone drains.

*Why overhead irrigation adds no transpiration.* Water intercepted by the canopy evaporates with the energy that would otherwise drive transpiration, so the canopy transpires less while it is wet. The intercepted evaporation replaces transpiration rather than adding to it, and the marginal transpiration is nil or slightly negative.

*Why the dose scale does not depend on the soil.* The water that a dose can lose to drainage is the part that the crop and the surface do not use before it redistributes below the roots. The crop uses about one day of evapotranspiration in that time whatever the soil, so the dose scale is set by the atmosphere and the crop and not by the soil.

*Why the plateau depends on the soil.* Once a dose is large, the share that drains is the share of the excess that the soil passes on before the crop uses it, a race between drainage and uptake. Coarse sand passes water fast (high conductivity at field capacity), fine soils hold it, so the plateau is highest in coarse sand.

*Why early-season irrigation is more productive.* Early in the season the canopy is growing and its demand rises, so applied water is transpired; late in the season the cereal ripens and is harvested, and the same water is evaporated from bare soil or drains.

## S7. The data

Table S2 lists the 17 experiments of the data folder and Fig. S1 shows their outputs. The tables that follow are the result files of the analysis, printed in full or up to 220 rows; every one is also a sheet of the workbook `Data/Manuscript_Data/SWAFL_Manuscripts_Data.xlsx`, and every run is in `Data/Comprehensive_Summary`.

**Table S2:** The data of this study: the 17 Daisy experiments of `Data/Comprehensive_Summary` (10190 runs, 10041 valid; 20 years, 1980–1999, of Taastrup weather). Means of the valid runs; $Y$ yield (Mg ha$^{-1}$), $I$ irrigation, $D$ drainage at 1 m and $ET$ (mm yr$^{-1}$), $N_L$ nitrate leached (kg N ha$^{-1}$ yr$^{-1}$), $t$ run time of Daisy (s). Every row is one file of the folder, linked.

| Exp. | File | Runs | Valid | Factors varied | $Y$ | $I$ | $D$ | $ET$ | $N_L$ | $t$ |
|---|---|---|---|---|---|---|---|---|---|---|
| E0 | `Baselines` | 10 | 10 | soil | 2.19 | 0 | 55 | 418 | 158 | 231 |
| E1 | `Doses` | 200 | 189 | soil, rate, dose | 5.13 | 198 | 98 | 564 | 79 | 195 |
| E2 | `TriggerPressure` | 200 | 197 | soil, trigger | 4.70 | 169 | 80 | 559 | 90 | 139 |
| E3 | `SensorDepth` | 200 | 200 | soil, sensor depth | 4.94 | 199 | 86 | 583 | 86 | 146 |
| E4 | `Methods` | 100 | 96 | soil, method | 5.01 | 164 | 88 | 542 | 83 | 147 |
| E5 | `SHP Dynamics` | 300 | 300 | soil, method, tillage | 4.13 | 152 | 43 | 579 | 84 | 189 |
| E6 | `Mediterranean Pollution` | 210 | 193 | soil, method, trigger, N rate | 3.64 | 84 | 239 | 620 | 62 | 118 |
| E7 | `Mulching` | 200 | 200 | soil, mulch | 5.29 | 183 | 96 | 555 | 74 | 161 |
| E8 | `Rain Intensity Moisture` | 1000 | 1000 | initial suction, rain scale | 0.68 | 0 | -1 | 282 | -0 | 6 |
| E9 | `Soil Moisture Rain` | 1000 | 987 | soil, initial suction, rain scale | 0.75 | 0 | 19 | 247 | 33 | 7 |
| E10 | `Amount Timing Soil` | 1000 | 954 | soil, rate, dose, window | 4.42 | 139 | 76 | 532 | 97 | 146 |
| E11 | `Temp Evap Rain` | 1000 | 1000 | rain scale, warming, demand | 3.81 | 324 | 18 | 754 | 66 | 151 |
| E12 | `Sensor Trigger Rain` | 1000 | 1000 | trigger, sensor depth, rain scale | 5.88 | 260 | 35 | 670 | 32 | 155 |
| E13 | `N Water Soil` | 1000 | 975 | soil, method, rate, dose, trigger, N rate | 4.00 | 162 | 83 | 547 | 79 | 207 |
| E14 | `Mulch Rain` | 1000 | 1000 | rain scale, mulch | 6.15 | 235 | 43 | 634 | 25 | 173 |
| E15 | `Method Depth Amount` | 1000 | 1000 | method, rate, dose, root depth | 6.02 | 217 | 61 | 619 | 37 | 157 |
| E16 | `Edge Cases` | 770 | 740 | soil, method, rate, dose, trigger, sensor depth, rain scale, warming, N rate, demand | 2.17 | 344 | 185 | 600 | 93 | 224 |

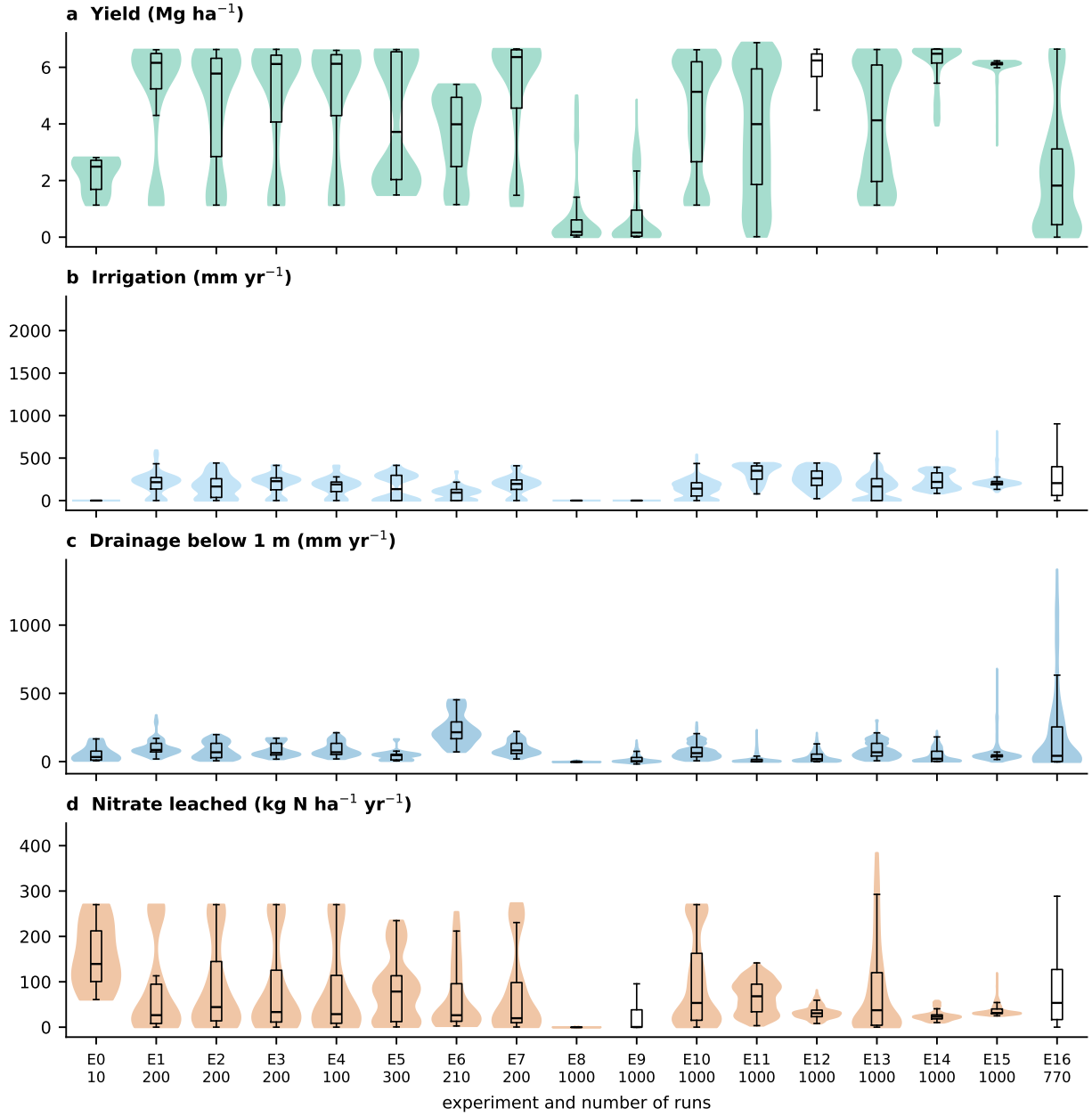



**Figure S1:** Yield, irrigation, drainage and nitrate leaching of the valid runs, by experiment.

## p4_tableA1_reach.csv

Source: `Manuscripts/results/p4_tableA1_reach.csv`; 10 rows. Workbook sheet `R_p4_tableA1_reach`.

| id | soil | alpha_1_per_cm | n | TAW_mm | reach_observed_low_cm | reach_observed_high_cm | reach_Eq4_cm | yield_at_300 |
|---|---|---|---|---|---|---|---|---|
| S01 | Coarse loam | 0.0961 | 1.353 | 130 | 1e+04 | inf | 7837 | 6.158 |
| S02 | Coarse sand | 0.0536 | 1.6 | 105 | 1e+04 | inf | 5373 | 6.254 |
| S03 | Sand | 0.145 | 2.68 | 60 | 100 | 150 | 199 | 1.132 |
| S04 | Loamy sand | 0.124 | 2.28 | 75 | 200 | 250 | 425.9 | 1.378 |
| S05 | Sandy loam | 0.075 | 1.89 | 95 | 800 | 1000 | 1618 | 4.351 |
| S06 | Loam | 0.036 | 1.56 | 150 | 5000 | 1e+04 | 9209 | 6.123 |
| S07 | Silt loam | 0.02 | 1.41 | 220 | 1e+04 | inf | 2.961e+04 | 6.573 |

| id | soil | alpha_1_per_cm | n | TAW_mm | reach_observed_low_cm | reach_observed_high_cm | reach_Eq4_cm | yield_at_300 |
|---|---|---|---|---|---|---|---|---|
| S08 | Sandy clay loam | 0.059 | 1.48 | 160 | 5000 | 1e+04 | 7580 | 5.853 |
| S09 | Clay loam | 0.019 | 1.31 | 175 | 1e+04 | inf | 4.776e+04 | 6.584 |
| S10 | Silty clay loam | 0.01 | 1.23 | 195 | 1e+04 | inf | 1.283e+05 | 6.575 |

## p4_table2_laws.csv

Source: `Manuscripts/results/p4_table2_laws.csv`; 3 rows. Workbook sheet `R_p4_table2_laws`.

| law | theta | loso_rmse_log10 | inside_of_10 | worst_factor |
|---|---|---|---|---|
| mualem | -9.46 | 0.1753 | 5 | 2.3 |
| power | 1.832 3.12 | 0.1431 | 4 | 2.368 |
| Sc | -1.489 | 0.2303 | 4 | 3.217 |

## p4_fig4a_fate_by_soil.csv

Source: `Manuscripts/results/p4_fig4a_fate_by_soil.csv`; 8 rows. Workbook sheet `R_p4_fig4a_fate_by_soil`.

| soil | transpired | soil_evaporation | other_evaporation | drained | storage_runoff |
|---|---|---|---|---|---|
| S01 | 0.3801 | 0.3228 | 0.1863 | 0.08046 | 0.03034 |
| S02 | 0.2323 | 0.1865 | 0.2611 | 0.3161 | 0.00396 |
| S05 | 0.6375 | 0.113 | 0.1369 | 0.07473 | 0.03789 |
| S06 | 0.4093 | 0.2202 | 0.1739 | 0.1519 | 0.04473 |
| S07 | 0.3523 | 0.2951 | 0.187 | 0.1478 | 0.01774 |
| S08 | 0.4586 | 0.1843 | 0.1643 | 0.1474 | 0.04543 |
| S09 | 0.3626 | 0.2869 | 0.1812 | 0.1529 | 0.01649 |
| S10 | 0.3406 | 0.292 | 0.1772 | 0.1466 | 0.04357 |

## p4_fig4b_fate_by_method.csv

Source: `Manuscripts/results/p4_fig4b_fate_by_method.csv`; 6 rows. Workbook sheet `R_p4_fig4b_fate_by_method`.

| experiment | method | transpired | soil_evaporation | other_evaporation | drained |
|---|---|---|---|---|---|
| E4 | overhead | -0.08 | 0.117 | 0.786 | 0.144 |
| E4 | subsoil | 0.511 | 0.177 | 0.04 | 0.228 |
| E4 | surface | 0.336 | 0.068 | 0.428 | 0.138 |
| E15 | overhead | -0.01 | 0.274 | 0.559 | 0.13 |
| E15 | subsoil | 0.404 | 0.234 | 0.033 | 0.282 |
| E15 | surface | 0.325 | 0.255 | 0.242 | 0.131 |

## p4_window.csv

Source: `Manuscripts/results/p4_window.csv`; 10 rows. Workbook sheet `R_p4_window`.

| window | etaT | etaD | I |
|---|---|---|---|
| 4/1-10/1 | 0.344 | 0.209 | 1.85e+04 |
| 4/1-6/1 | 0.641 | 0.072 | 7917 |
| 4/1-7/1 | 0.477 | 0.153 | 1.262e+04 |
| 4/15-10/1 | 0.352 | 0.19 | 1.714e+04 |
| 5/1-10/1 | 0.348 | 0.164 | 1.592e+04 |
| 5/1-7/15 | 0.431 | 0.129 | 1.219e+04 |
| 5/15-10/1 | 0.34 | 0.157 | 1.538e+04 |
| 6/1-10/1 | 0.326 | 0.148 | 1.328e+04 |
| 6/1-8/15 | 0.362 | 0.119 | 1.097e+04 |
| 7/1-9/15 | 0.234 | 0.136 | 8153 |

## p4_fig5_dose.csv

Source: `Manuscripts/results/p4_fig5_dose.csv`; 77 rows. Workbook sheet `R_p4_fig5_dose`.

| soil | dose_mm | drained | transpired | evaporated | fit_plateau | fit_dose_scale_mm | fit_R2 |
|---|---|---|---|---|---|---|---|
| S01 | 1 | 0.01116 | 0.2724 | 0.7167 | 0.1094 | 4.425 | 0.9492 |
| S01 | 2 | 0.02367 | 0.3536 | 0.6117 | 0.1094 | 4.425 | 0.9492 |
| S01 | 3 | 0.05371 | 0.3646 | 0.553 | 0.1094 | 4.425 | 0.9492 |
| S01 | 4 | 0.07486 | 0.3772 | 0.5149 | 0.1094 | 4.425 | 0.9492 |
| S01 | 6 | 0.08865 | 0.3887 | 0.4882 | 0.1094 | 4.425 | 0.9492 |
| S01 | 8 | 0.09569 | 0.3917 | 0.4783 | 0.1094 | 4.425 | 0.9492 |
| S01 | 10 | 0.09623 | 0.3973 | 0.4726 | 0.1094 | 4.425 | 0.9492 |
| S01 | 15 | 0.1041 | 0.3972 | 0.464 | 0.1094 | 4.425 | 0.9492 |
| S01 | 20 | 0.1051 | 0.4052 | 0.4561 | 0.1094 | 4.425 | 0.9492 |
| S01 | 30 | 0.1088 | 0.3969 | 0.452 | 0.1094 | 4.425 | 0.9492 |
| S02 | 1 | 0.09002 | 0.1999 | 0.707 | 0.4448 | 3.234 | 0.9548 |
| S02 | 2 | 0.1722 | 0.2894 | 0.5354 | 0.4448 | 3.234 | 0.9548 |
| S02 | 3 | 0.2796 | 0.2777 | 0.4387 | 0.4448 | 3.234 | 0.9548 |
| S02 | 4 | 0.3611 | 0.2207 | 0.4135 | 0.4448 | 3.234 | 0.9548 |
| S02 | 6 | 0.3737 | 0.2353 | 0.3861 | 0.4448 | 3.234 | 0.9548 |
| S02 | 8 | 0.4047 | 0.182 | 0.41 | 0.4448 | 3.234 | 0.9548 |
| S02 | 10 | 0.4119 | 0.1804 | 0.4044 | 0.4448 | 3.234 | 0.9548 |
| S05 | 1 | 0.02737 | 0.3501 | 0.6058 | 0.1017 | 5.21 | 0.6575 |
| S05 | 2 | 0.04118 | 0.5204 | 0.4112 | 0.1017 | 5.21 | 0.6575 |
| S05 | 3 | 0.05739 | 0.5942 | 0.3121 | 0.1017 | 5.21 | 0.6575 |
| S05 | 4 | 0.06418 | 0.6268 | 0.2689 | 0.1017 | 5.21 | 0.6575 |
| S05 | 6 | 0.07016 | 0.6572 | 0.2316 | 0.1017 | 5.21 | 0.6575 |
| S05 | 8 | 0.06801 | 0.6813 | 0.2103 | 0.1017 | 5.21 | 0.6575 |
| S05 | 10 | 0.07391 | 0.6866 | 0.1991 | 0.1017 | 5.21 | 0.6575 |
| S05 | 15 | 0.06929 | 0.7145 | 0.1781 | 0.1017 | 5.21 | 0.6575 |

| soil | dose_mm | drained | transpired | evaporated | fit_plateau | fit_dose_scale_mm | fit_R2 |
|---|---|---|---|---|---|---|---|
| S05 | 20 | 0.09447 | 0.695 | 0.1704 | 0.1017 | 5.21 | 0.6575 |
| S05 | 30 | 0.1362 | 0.6673 | 0.1531 | 0.1017 | 5.21 | 0.6575 |
| S06 | 1 | 0.01796 | 0.2696 | 0.7016 | 0.1971 | 3.887 | 0.9311 |
| S06 | 2 | 0.06301 | 0.3512 | 0.5581 | 0.1971 | 3.887 | 0.9311 |
| S06 | 3 | 0.1161 | 0.3796 | 0.4601 | 0.1971 | 3.887 | 0.9311 |
| S06 | 4 | 0.1462 | 0.4004 | 0.4051 | 0.1971 | 3.887 | 0.9311 |
| S06 | 6 | 0.165 | 0.4219 | 0.3629 | 0.1971 | 3.887 | 0.9311 |
| S06 | 8 | 0.1693 | 0.432 | 0.3484 | 0.1971 | 3.887 | 0.9311 |
| S06 | 10 | 0.1729 | 0.4367 | 0.3402 | 0.1971 | 3.887 | 0.9311 |
| S06 | 15 | 0.1805 | 0.4459 | 0.3234 | 0.1971 | 3.887 | 0.9311 |
| S06 | 20 | 0.1857 | 0.4441 | 0.3205 | 0.1971 | 3.887 | 0.9311 |
| S06 | 30 | 0.2169 | 0.4317 | 0.3053 | 0.1971 | 3.887 | 0.9311 |
| S07 | 1 | 0.01468 | 0.2724 | 0.7147 | 0.194 | 4.106 | 0.9292 |
| S07 | 2 | 0.05912 | 0.3413 | 0.5963 | 0.194 | 4.106 | 0.9292 |
| S07 | 3 | 0.1092 | 0.3436 | 0.5296 | 0.194 | 4.106 | 0.9292 |
| S07 | 4 | 0.1408 | 0.3477 | 0.4905 | 0.194 | 4.106 | 0.9292 |
| S07 | 6 | 0.1589 | 0.3575 | 0.4622 | 0.194 | 4.106 | 0.9292 |
| S07 | 8 | 0.1641 | 0.3623 | 0.4525 | 0.194 | 4.106 | 0.9292 |
| S07 | 10 | 0.1699 | 0.3642 | 0.445 | 0.194 | 4.106 | 0.9292 |
| S07 | 15 | 0.174 | 0.3671 | 0.4382 | 0.194 | 4.106 | 0.9292 |
| S07 | 20 | 0.1823 | 0.366 | 0.4307 | 0.194 | 4.106 | 0.9292 |
| S07 | 30 | 0.2145 | 0.3564 | 0.4098 | 0.194 | 4.106 | 0.9292 |
| S08 | 1 | 0.0296 | 0.2739 | 0.6874 | 0.19 | 3.85 | 0.9452 |
| S08 | 2 | 0.06499 | 0.3832 | 0.5247 | 0.19 | 3.85 | 0.9452 |
| S08 | 3 | 0.1097 | 0.4284 | 0.4178 | 0.19 | 3.85 | 0.9452 |
| S08 | 4 | 0.1397 | 0.4525 | 0.3585 | 0.19 | 3.85 | 0.9452 |
| S08 | 6 | 0.1563 | 0.4748 | 0.3173 | 0.19 | 3.85 | 0.9452 |
| S08 | 8 | 0.1613 | 0.4865 | 0.3006 | 0.19 | 3.85 | 0.9452 |
| S08 | 10 | 0.1666 | 0.4892 | 0.2926 | 0.19 | 3.85 | 0.9452 |
| S08 | 15 | 0.1741 | 0.4977 | 0.2767 | 0.19 | 3.85 | 0.9452 |
| S08 | 20 | 0.1798 | 0.5015 | 0.2683 | 0.19 | 3.85 | 0.9452 |
| S08 | 30 | 0.2122 | 0.4892 | 0.2521 | 0.19 | 3.85 | 0.9452 |
| S09 | 1 | 0.01645 | 0.2722 | 0.7152 | 0.2043 | 4.388 | 0.926 |
| S09 | 2 | 0.06101 | 0.3491 | 0.5921 | 0.2043 | 4.388 | 0.926 |
| S09 | 3 | 0.1096 | 0.3522 | 0.523 | 0.2043 | 4.388 | 0.926 |
| S09 | 4 | 0.1438 | 0.3592 | 0.4765 | 0.2043 | 4.388 | 0.926 |
| S09 | 6 | 0.1617 | 0.3707 | 0.4464 | 0.2043 | 4.388 | 0.926 |
| S09 | 8 | 0.1695 | 0.3732 | 0.4365 | 0.2043 | 4.388 | 0.926 |
| S09 | 10 | 0.173 | 0.3759 | 0.43 | 0.2043 | 4.388 | 0.926 |
| S09 | 15 | 0.1765 | 0.3798 | 0.4228 | 0.2043 | 4.388 | 0.926 |
| S09 | 20 | 0.1938 | 0.3783 | 0.4081 | 0.2043 | 4.388 | 0.926 |
| S09 | 30 | 0.2299 | 0.3654 | 0.387 | 0.2043 | 4.388 | 0.926 |
| S10 | 1 | 0.01687 | 0.2859 | 0.7031 | 0.1827 | 3.394 | 0.923 |
| S10 | 2 | 0.06202 | 0.3556 | 0.5896 | 0.1827 | 3.394 | 0.923 |
| S10 | 3 | 0.1119 | 0.3491 | 0.528 | 0.1827 | 3.394 | 0.923 |
| S10 | 4 | 0.1439 | 0.3487 | 0.4899 | 0.1827 | 3.394 | 0.923 |
| S10 | 6 | 0.1635 | 0.3542 | 0.4635 | 0.1827 | 3.394 | 0.923 |
| S10 | 8 | 0.1683 | 0.3576 | 0.4561 | 0.1827 | 3.394 | 0.923 |

| soil | dose_mm | drained | transpired | evaporated | fit_plateau | fit_dose_scale_mm | fit_R2 |
|---|---|---|---|---|---|---|---|
| S10 | 10 | 0.1773 | 0.3561 | 0.4482 | 0.1827 | 3.394 | 0.923 |
| S10 | 15 | 0.176 | 0.3599 | 0.4442 | 0.1827 | 3.394 | 0.923 |
| S10 | 20 | 0.1911 | 0.3542 | 0.427 | 0.1827 | 3.394 | 0.923 |
| S10 | 30 | 0.1629 | 0.2778 | 0.3331 | 0.1827 | 3.394 | 0.923 |

## p4_fig3_trigger_runs.csv

Source: `Manuscripts/results/p4_fig3_trigger_runs.csv`; 197 rows. Workbook sheet `R_p4_fig3_trigger_runs`.

| soil | trigger_cm | irrigation_mm | yield_Mg_ha | trigger_over_reach | Inorm |
|---|---|---|---|---|---|
| S01_CoarseLoam | 100 | 296.9 | 6.62 | 0.01276 | 0.8858 |
| S01_CoarseLoam | 1000 | 168.7 | 5.928 | 0.1276 | 0.5034 |
| S01_CoarseLoam | 1e+04 | 60.48 | 3.798 | 1.276 | 0.1805 |
| S01_CoarseLoam | 1200 | 160.8 | 5.828 | 0.1531 | 0.4798 |
| S01_CoarseLoam | 150 | 270.8 | 6.317 | 0.01914 | 0.8081 |
| S01_CoarseLoam | 1500 | 152.3 | 5.707 | 0.1914 | 0.4544 |
| S01_CoarseLoam | 200 | 254 | 6.271 | 0.02552 | 0.758 |
| S01_CoarseLoam | 2000 | 142.1 | 5.526 | 0.2552 | 0.4239 |
| S01_CoarseLoam | 250 | 242.5 | 6.318 | 0.0319 | 0.7236 |
| S01_CoarseLoam | 300 | 228.5 | 6.158 | 0.03828 | 0.6817 |
| S01_CoarseLoam | 3000 | 124.1 | 5.087 | 0.3828 | 0.3702 |
| S01_CoarseLoam | 350 | 219 | 6.088 | 0.04466 | 0.6534 |
| S01_CoarseLoam | 400 | 211.8 | 6.078 | 0.05104 | 0.6319 |
| S01_CoarseLoam | 50 | 335.2 | 6.62 | 0.00638 | 1 |
| S01_CoarseLoam | 500 | 202.3 | 6.064 | 0.0638 | 0.6037 |
| S01_CoarseLoam | 5000 | 97.92 | 4.547 | 0.638 | 0.2922 |
| S01_CoarseLoam | 600 | 191.5 | 6.041 | 0.07656 | 0.5714 |
| S01_CoarseLoam | 700 | 183.8 | 6.022 | 0.08932 | 0.5485 |
| S01_CoarseLoam | 75 | 311.8 | 6.624 | 0.00957 | 0.9302 |
| S01_CoarseLoam | 800 | 177.5 | 5.996 | 0.1021 | 0.5295 |
| S02_CoarseSand | 100 | 438.8 | 6.317 | 0.01861 | 0.9938 |
| S02_CoarseSand | 1e+04 | 139.3 | 3.853 | 1.861 | 0.3155 |
| S02_CoarseSand | 150 | 433.2 | 6.306 | 0.02792 | 0.981 |
| S02_CoarseSand | 200 | 427.1 | 6.295 | 0.03723 | 0.9671 |
| S02_CoarseSand | 2000 | 248 | 4.861 | 0.3723 | 0.5617 |
| S02_CoarseSand | 250 | 419.4 | 6.281 | 0.04653 | 0.9497 |
| S02_CoarseSand | 300 | 411.1 | 6.254 | 0.05584 | 0.931 |
| S02_CoarseSand | 3000 | 218.5 | 4.586 | 0.5584 | 0.4948 |
| S02_CoarseSand | 350 | 402.6 | 6.222 | 0.06514 | 0.9117 |
| S02_CoarseSand | 400 | 394.1 | 6.189 | 0.07445 | 0.8924 |
| S02_CoarseSand | 50 | 441.6 | 6.32 | 0.009306 | 1 |
| S02_CoarseSand | 500 | 379.2 | 6.106 | 0.09306 | 0.8587 |
| S02_CoarseSand | 5000 | 180.7 | 4.283 | 0.9306 | 0.4092 |
| S02_CoarseSand | 600 | 366.4 | 6.044 | 0.1117 | 0.8296 |

| soil | trigger_cm | irrigation_mm | yield_Mg_ha | trigger_over_reach | Inorm |
|---|---|---|---|---|---|
| S02_CoarseSand | 700 | 352.4 | 5.916 | 0.1303 | 0.7981 |
| S02_CoarseSand | 75 | 441.2 | 6.32 | 0.01396 | 0.9992 |
| S02_CoarseSand | 800 | 341 | 5.844 | 0.1489 | 0.7723 |
| S03_Sand | 100 | 6.36 | 1.234 | 0.5024 | 0.05206 |
| S03_Sand | 1000 | 0 | 1.132 | 5.024 | 0 |
| S03_Sand | 1e+04 | 0 | 1.132 | 50.24 | 0 |
| S03_Sand | 1200 | 0 | 1.132 | 6.029 | 0 |
| S03_Sand | 150 | 0 | 1.132 | 0.7536 | 0 |
| S03_Sand | 1500 | 0 | 1.132 | 7.536 | 0 |
| S03_Sand | 200 | 0 | 1.132 | 1.005 | 0 |
| S03_Sand | 2000 | 0 | 1.132 | 10.05 | 0 |
| S03_Sand | 250 | 0 | 1.132 | 1.256 | 0 |
| S03_Sand | 300 | 0 | 1.132 | 1.507 | 0 |
| S03_Sand | 3000 | 0 | 1.132 | 15.07 | 0 |
| S03_Sand | 350 | 0 | 1.132 | 1.758 | 0 |
| S03_Sand | 400 | 0 | 1.132 | 2.01 | 0 |
| S03_Sand | 50 | 122.2 | 3.038 | 0.2512 | 1 |
| S03_Sand | 500 | 0 | 1.132 | 2.512 | 0 |
| S03_Sand | 5000 | 0 | 1.132 | 25.12 | 0 |
| S03_Sand | 600 | 0 | 1.132 | 3.014 | 0 |
| S03_Sand | 700 | 0 | 1.132 | 3.517 | 0 |
| S03_Sand | 75 | 38.52 | 1.765 | 0.3768 | 0.3153 |
| S03_Sand | 800 | 0 | 1.132 | 4.019 | 0 |
| S04_LoamySand | 100 | 122.9 | 3.455 | 0.2348 | 0.5508 |
| S04_LoamySand | 1000 | 0 | 1.378 | 2.348 | 0 |
| S04_LoamySand | 1e+04 | 0 | 1.378 | 23.48 | 0 |
| S04_LoamySand | 1200 | 0 | 1.378 | 2.818 | 0 |
| S04_LoamySand | 150 | 50.04 | 2.276 | 0.3522 | 0.2243 |
| S04_LoamySand | 1500 | 0 | 1.378 | 3.522 | 0 |
| S04_LoamySand | 200 | 16.8 | 1.67 | 0.4696 | 0.07531 |
| S04_LoamySand | 2000 | 0 | 1.378 | 4.696 | 0 |
| S04_LoamySand | 250 | 4.08 | 1.433 | 0.587 | 0.01829 |
| S04_LoamySand | 300 | 0 | 1.378 | 0.7044 | 0 |
| S04_LoamySand | 3000 | 0 | 1.378 | 7.044 | 0 |
| S04_LoamySand | 350 | 0 | 1.378 | 0.8218 | 0 |
| S04_LoamySand | 400 | 0 | 1.378 | 0.9392 | 0 |
| S04_LoamySand | 50 | 223.1 | 4.544 | 0.1174 | 1 |
| S04_LoamySand | 500 | 0 | 1.378 | 1.174 | 0 |
| S04_LoamySand | 5000 | 0 | 1.378 | 11.74 | 0 |
| S04_LoamySand | 600 | 0 | 1.378 | 1.409 | 0 |
| S04_LoamySand | 700 | 0 | 1.378 | 1.644 | 0 |
| S04_LoamySand | 75 | 152.8 | 3.893 | 0.1761 | 0.6848 |
| S04_LoamySand | 800 | 0 | 1.378 | 1.878 | 0 |
| S05_SandyLoam | 100 | 230 | 5.263 | 0.06179 | 0.6291 |
| S05_SandyLoam | 1000 | 4.08 | 1.986 | 0.6179 | 0.01116 |
| S05_SandyLoam | 1e+04 | 0 | 1.851 | 6.179 | 0 |
| S05_SandyLoam | 1200 | 0 | 1.851 | 0.7415 | 0 |
| S05_SandyLoam | 150 | 181.7 | 5.016 | 0.09268 | 0.4969 |

| soil | trigger_cm | irrigation_mm | yield_Mg_ha | trigger_over_reach | Inorm |
|---|---|---|---|---|---|
| S05_SandyLoam | 1500 | 0 | 1.851 | 0.9268 | 0 |
| S05_SandyLoam | 200 | 159.4 | 4.783 | 0.1236 | 0.4358 |
| S05_SandyLoam | 2000 | 0 | 1.851 | 1.236 | 0 |
| S05_SandyLoam | 250 | 143.3 | 4.579 | 0.1545 | 0.3919 |
| S05_SandyLoam | 300 | 127.6 | 4.351 | 0.1854 | 0.3489 |
| S05_SandyLoam | 3000 | 0 | 1.851 | 1.854 | 0 |
| S05_SandyLoam | 350 | 117.1 | 4.094 | 0.2163 | 0.3203 |
| S05_SandyLoam | 400 | 99.12 | 3.834 | 0.2472 | 0.2711 |
| S05_SandyLoam | 50 | 365.6 | 6.02 | 0.03089 | 1 |
| S05_SandyLoam | 500 | 72.12 | 3.314 | 0.3089 | 0.1972 |
| S05_SandyLoam | 5000 | 0 | 1.851 | 3.089 | 0 |
| S05_SandyLoam | 600 | 47.76 | 2.847 | 0.3707 | 0.1306 |
| S05_SandyLoam | 700 | 35.88 | 2.652 | 0.4325 | 0.09813 |
| S05_SandyLoam | 75 | 282.7 | 5.756 | 0.04634 | 0.7732 |
| S05_SandyLoam | 800 | 17.52 | 2.28 | 0.4943 | 0.04792 |
| S06_Loam | 100 | 346.4 | 6.606 | 0.01086 | 0.8178 |
| S06_Loam | 1000 | 116.2 | 5.189 | 0.1086 | 0.2742 |
| S06_Loam | 1e+04 | 0 | 2.615 | 1.086 | 0 |
| S06_Loam | 1200 | 104.6 | 4.909 | 0.1303 | 0.247 |
| S06_Loam | 150 | 294 | 6.552 | 0.01629 | 0.6941 |
| S06_Loam | 1500 | 87.12 | 4.597 | 0.1629 | 0.2057 |
| S06_Loam | 200 | 257.3 | 6.385 | 0.02172 | 0.6074 |
| S06_Loam | 2000 | 67.8 | 4.199 | 0.2172 | 0.1601 |
| S06_Loam | 250 | 230.3 | 6.161 | 0.02715 | 0.5436 |
| S06_Loam | 300 | 211.7 | 6.123 | 0.03258 | 0.4997 |
| S06_Loam | 3000 | 36.72 | 3.574 | 0.3258 | 0.08669 |
| S06_Loam | 350 | 201 | 6.098 | 0.038 | 0.4745 |
| S06_Loam | 400 | 189.6 | 6.063 | 0.04343 | 0.4476 |
| S06_Loam | 50 | 423.6 | 6.623 | 0.005429 | 1 |
| S06_Loam | 500 | 173 | 5.955 | 0.05429 | 0.4085 |
| S06_Loam | 5000 | 5.16 | 2.764 | 0.5429 | 0.01218 |
| S06_Loam | 600 | 158.8 | 5.806 | 0.06515 | 0.3748 |
| S06_Loam | 700 | 147.1 | 5.672 | 0.07601 | 0.3473 |
| S06_Loam | 75 | 381.4 | 6.62 | 0.008144 | 0.9003 |
| S06_Loam | 800 | 137.4 | 5.523 | 0.08687 | 0.3244 |
| S07_SiltLoam | 100 | 379.1 | 6.624 | 0.003377 | 0.8797 |
| S07_SiltLoam | 1000 | 166.9 | 6.198 | 0.03377 | 0.3874 |
| S07_SiltLoam | 1e+04 | 19.2 | 3.341 | 0.3377 | 0.04456 |
| S07_SiltLoam | 1200 | 157 | 6.06 | 0.04052 | 0.3642 |
| S07_SiltLoam | 150 | 338.5 | 6.622 | 0.005065 | 0.7856 |
| S07_SiltLoam | 1500 | 145.4 | 5.838 | 0.05065 | 0.3375 |
| S07_SiltLoam | 200 | 311.6 | 6.62 | 0.006754 | 0.7232 |
| S07_SiltLoam | 2000 | 125.5 | 5.529 | 0.06754 | 0.2913 |
| S07_SiltLoam | 250 | 290.6 | 6.628 | 0.008442 | 0.6745 |
| S07_SiltLoam | 300 | 271.8 | 6.573 | 0.01013 | 0.6307 |
| S07_SiltLoam | 3000 | 107.8 | 5.178 | 0.1013 | 0.2501 |
| S07_SiltLoam | 350 | 257.5 | 6.564 | 0.01182 | 0.5976 |
| S07_SiltLoam | 400 | 242.8 | 6.425 | 0.01351 | 0.5634 |

| soil | trigger_cm | irrigation_mm | yield_Mg_ha | trigger_over_reach | Inorm |
|---|---|---|---|---|---|
| S07_SiltLoam | 50 | 430.9 | 6.626 | 0.001688 | 1 |
| S07_SiltLoam | 500 | 218.6 | 6.388 | 0.01688 | 0.5074 |
| S07_SiltLoam | 5000 | 79.68 | 4.654 | 0.1688 | 0.1849 |
| S07_SiltLoam | 600 | 199.9 | 6.359 | 0.02026 | 0.4639 |
| S07_SiltLoam | 700 | 188.4 | 6.31 | 0.02364 | 0.4372 |
| S07_SiltLoam | 75 | 406.2 | 6.623 | 0.002533 | 0.9426 |
| S07_SiltLoam | 800 | 180.6 | 6.288 | 0.02702 | 0.4191 |
| S08_SandyClayLoam | 100 | 294.5 | 6.449 | 0.01319 | 0.7805 |
| S08_SandyClayLoam | 1000 | 109.3 | 4.773 | 0.1319 | 0.2898 |
| S08_SandyClayLoam | 1e+04 | 0 | 2.384 | 1.319 | 0 |
| S08_SandyClayLoam | 1200 | 97.2 | 4.527 | 0.1583 | 0.2576 |
| S08_SandyClayLoam | 150 | 247.2 | 6.144 | 0.01979 | 0.6552 |
| S08_SandyClayLoam | 1500 | 82.2 | 4.276 | 0.1979 | 0.2179 |
| S08_SandyClayLoam | 200 | 220.2 | 5.998 | 0.02639 | 0.5837 |
| S08_SandyClayLoam | 2000 | 64.2 | 3.923 | 0.2639 | 0.1702 |
| S08_SandyClayLoam | 250 | 205.4 | 5.91 | 0.03298 | 0.5445 |
| S08_SandyClayLoam | 300 | 191.5 | 5.853 | 0.03958 | 0.5076 |
| S08_SandyClayLoam | 3000 | 36.84 | 3.348 | 0.3958 | 0.09765 |
| S08_SandyClayLoam | 350 | 180.5 | 5.79 | 0.04617 | 0.4784 |
| S08_SandyClayLoam | 400 | 171.2 | 5.725 | 0.05277 | 0.4539 |
| S08_SandyClayLoam | 50 | 377.3 | 6.566 | 0.006596 | 1 |
| S08_SandyClayLoam | 500 | 158.5 | 5.554 | 0.06596 | 0.4202 |
| S08_SandyClayLoam | 5000 | 6 | 2.546 | 0.6596 | 0.0159 |
| S08_SandyClayLoam | 600 | 146.5 | 5.399 | 0.07916 | 0.3884 |
| S08_SandyClayLoam | 700 | 134.3 | 5.229 | 0.09235 | 0.3559 |
| S08_SandyClayLoam | 75 | 330.2 | 6.515 | 0.009895 | 0.8753 |
| S08_SandyClayLoam | 800 | 121.3 | 5.039 | 0.1055 | 0.3216 |
| S09_ClayLoam | 100 | 354.8 | 6.617 | 0.002094 | 0.8718 |
| S09_ClayLoam | 1000 | 174.5 | 6.311 | 0.02094 | 0.4287 |
| S09_ClayLoam | 1e+04 | 60.72 | 4.275 | 0.2094 | 0.1492 |
| S09_ClayLoam | 1200 | 165.5 | 6.21 | 0.02513 | 0.4065 |
| S09_ClayLoam | 150 | 324.1 | 6.618 | 0.003141 | 0.7963 |
| S09_ClayLoam | 1500 | 157.7 | 6.031 | 0.03141 | 0.3874 |
| S09_ClayLoam | 200 | 300.6 | 6.62 | 0.004188 | 0.7385 |
| S09_ClayLoam | 2000 | 143.6 | 5.778 | 0.04188 | 0.3529 |
| S09_ClayLoam | 250 | 282.5 | 6.622 | 0.005235 | 0.694 |
| S09_ClayLoam | 300 | 266.9 | 6.584 | 0.006282 | 0.6557 |
| S09_ClayLoam | 3000 | 124.3 | 5.436 | 0.06282 | 0.3054 |
| S09_ClayLoam | 350 | 253.7 | 6.55 | 0.007329 | 0.6232 |
| S09_ClayLoam | 400 | 241.6 | 6.474 | 0.008376 | 0.5935 |
| S09_ClayLoam | 50 | 407 | 6.605 | 0.001047 | 1 |
| S09_ClayLoam | 500 | 221.9 | 6.451 | 0.01047 | 0.5451 |
| S09_ClayLoam | 5000 | 105.5 | 5.079 | 0.1047 | 0.2591 |
| S09_ClayLoam | 600 | 205.2 | 6.427 | 0.01256 | 0.5041 |
| S09_ClayLoam | 700 | 193.9 | 6.409 | 0.01466 | 0.4764 |
| S09_ClayLoam | 75 | 377.6 | 6.62 | 0.00157 | 0.9278 |
| S09_ClayLoam | 800 | 185.8 | 6.386 | 0.01675 | 0.4564 |
| S10_SiltyClayLoam | 100 | 346.8 | 6.229 | 0.0007797 | 0.9051 |

| soil | trigger_cm | irrigation_mm | yield_Mg_ha | trigger_over_reach | Inorm |
|---|---|---|---|---|---|
| S10_SiltyClayLoam | 1000 | 190.3 | 6.425 | 0.007797 | 0.4967 |
| S10_SiltyClayLoam | 1e+04 | 95.04 | 4.905 | 0.07797 | 0.248 |
| S10_SiltyClayLoam | 1200 | 182 | 6.393 | 0.009356 | 0.4751 |
| S10_SiltyClayLoam | 150 | 324.6 | 6.418 | 0.00117 | 0.8472 |
| S10_SiltyClayLoam | 1500 | 175.1 | 6.318 | 0.0117 | 0.4569 |
| S10_SiltyClayLoam | 200 | 306 | 6.509 | 0.001559 | 0.7986 |
| S10_SiltyClayLoam | 2000 | 163.1 | 6.163 | 0.01559 | 0.4256 |
| S10_SiltyClayLoam | 250 | 291.5 | 6.547 | 0.001949 | 0.7607 |
| S10_SiltyClayLoam | 300 | 278.4 | 6.575 | 0.002339 | 0.7266 |
| S10_SiltyClayLoam | 3000 | 149.9 | 5.874 | 0.02339 | 0.3912 |
| S10_SiltyClayLoam | 350 | 268.2 | 6.585 | 0.002729 | 0.7 |
| S10_SiltyClayLoam | 400 | 259.8 | 6.59 | 0.003119 | 0.678 |
| S10_SiltyClayLoam | 50 | 383.2 | 5.986 | 0.0003898 | 1 |
| S10_SiltyClayLoam | 500 | 241.6 | 6.507 | 0.003898 | 0.6304 |
| S10_SiltyClayLoam | 5000 | 127.9 | 5.496 | 0.03898 | 0.3339 |
| S10_SiltyClayLoam | 600 | 225.4 | 6.491 | 0.004678 | 0.5882 |
| S10_SiltyClayLoam | 700 | 212.6 | 6.469 | 0.005458 | 0.555 |
| S10_SiltyClayLoam | 75 | 362.6 | 6.103 | 0.0005848 | 0.9464 |
| S10_SiltyClayLoam | 800 | 203.4 | 6.455 | 0.006237 | 0.5308 |

## p4_numbers.json

Source: `Manuscripts/results/p4_numbers.json`; 295 values, first 220 shown. Workbook sheet `R_p4_numbers`.

| quantity | value |
|---|---|
| runs_total | 10190 |
| runs_valid | 10041 |
| E2_runs | 197 |
| reach.mualem.theta | -9.46 |
| reach.mualem.loso_err_log10 | 0.169, 0.362, 0.134, 0.26, 0.25, 0, 0, 0, 0, 0 |
| reach.mualem.loso_rms_log10 | 0.1753 |
| reach.mualem.loso_inside | 5 |
| reach.mualem.loso_max_factor | 2.3 |
| reach.mualem.pred.S01 | 7837 |
| reach.mualem.pred.S02 | 5373 |
| reach.mualem.pred.S03 | 199 |
| reach.mualem.pred.S04 | 425.9 |
| reach.mualem.pred.S05 | 1618 |
| reach.mualem.pred.S06 | 9209 |
| reach.mualem.pred.S07 | 2.961e+04 |
| reach.mualem.pred.S08 | 7580 |
| reach.mualem.pred.S09 | 4.776e+04 |
| reach.mualem.pred.S10 | 1.283e+05 |
| reach.power.theta | 1.832, 3.12 |

| quantity | value |
|---|---|
| reach.power.loso_err_log10 | 0, 0.374, 0.172, 0.009, 0.139, 0.086, 0, 0.092, 0, 0 |
| reach.power.loso_rms_log10 | 0.1431 |
| reach.power.loso_inside | 4 |
| reach.power.loso_max_factor | 2.368 |
| reach.power.pred.S01 | 1.82e+04 |
| reach.power.pred.S02 | 6235 |
| reach.power.pred.S03 | 92.83 |
| reach.power.pred.S04 | 253.5 |
| reach.power.pred.S05 | 1302 |
| reach.power.pred.S06 | 1.151e+04 |
| reach.power.pred.S07 | 5.481e+04 |
| reach.power.pred.S08 | 1.136e+04 |
| reach.power.pred.S09 | 1.38e+05 |
| reach.power.pred.S10 | 6.654e+05 |
| reach.Sc.theta | -1.489 |
| reach.Sc.loso_err_log10 | 0, 0.317, 0.284, 0.243, 0.117, 0.138, 0, 0.507, 0, 0 |
| reach.Sc.loso_rms_log10 | 0.2303 |
| reach.Sc.loso_inside | 4 |
| reach.Sc.loso_max_factor | 3.217 |
| reach.Sc.pred.S01 | 1.723e+05 |
| reach.Sc.pred.S02 | 5662 |
| reach.Sc.pred.S03 | 53.02 |
| reach.Sc.pred.S04 | 117.4 |
| reach.Sc.pred.S05 | 628.3 |
| reach.Sc.pred.S06 | 1.268e+04 |
| reach.Sc.pred.S07 | 2.145e+05 |
| reach.Sc.pred.S08 | 2.147e+04 |
| reach.Sc.pred.S09 | 3.353e+06 |
| reach.Sc.pred.S10 | 2.987e+08 |
| kappa | 3.47e-10 |
| blind_300 | S03, S04 |
| yield_E2.S01.rainfed | 2.597 |
| yield_E2.S01.best | 6.624 |
| yield_E2.S01.at300 | 6.158 |
| yield_E2.S02.rainfed | 1.633 |
| yield_E2.S02.best | 6.32 |
| yield_E2.S02.at300 | 6.254 |
| yield_E2.S03.rainfed | 1.132 |
| yield_E2.S03.best | 3.038 |
| yield_E2.S03.at300 | 1.132 |
| yield_E2.S04.rainfed | 1.378 |
| yield_E2.S04.best | 4.544 |
| yield_E2.S04.at300 | 1.378 |
| yield_E2.S05.rainfed | 1.851 |
| yield_E2.S05.best | 6.02 |
| yield_E2.S05.at300 | 4.351 |
| yield_E2.S06.rainfed | 2.615 |
| yield_E2.S06.best | 6.623 |

| quantity | value |
|---|---|
| yield_E2.S06.at300 | 6.123 |
| yield_E2.S07.rainfed | 2.813 |
| yield_E2.S07.best | 6.628 |
| yield_E2.S07.at300 | 6.573 |
| yield_E2.S08.rainfed | 2.384 |
| yield_E2.S08.best | 6.566 |
| yield_E2.S08.at300 | 5.853 |
| yield_E2.S09.rainfed | 2.773 |
| yield_E2.S09.best | 6.622 |
| yield_E2.S09.at300 | 6.584 |
| yield_E2.S10.rainfed | 2.75 |
| yield_E2.S10.best | 6.59 |
| yield_E2.S10.at300 | 6.575 |
| collapse_raw_h | 0.5407 |
| collapse_scaled | 0.3167 |
| spearman_scaled | -0.8339 |
| E10_runs | 954 |
| E10_runs_irrigated | 744 |
| fate_E10.S01.dT | 0.38 |
| fate_E10.S01.dE | 0.323 |
| fate_E10.S01.dO | 0.186 |
| fate_E10.S01.dD | 0.08 |
| fate_E10.S01.dS | 0.03 |
| fate_E10.S02.dT | 0.232 |
| fate_E10.S02.dE | 0.186 |
| fate_E10.S02.dO | 0.261 |
| fate_E10.S02.dD | 0.316 |
| fate_E10.S02.dS | 0.004 |
| fate_E10.S05.dT | 0.638 |
| fate_E10.S05.dE | 0.113 |
| fate_E10.S05.dO | 0.137 |
| fate_E10.S05.dD | 0.075 |
| fate_E10.S05.dS | 0.038 |
| fate_E10.S06.dT | 0.409 |
| fate_E10.S06.dE | 0.22 |
| fate_E10.S06.dO | 0.174 |
| fate_E10.S06.dD | 0.152 |
| fate_E10.S06.dS | 0.045 |
| fate_E10.S07.dT | 0.352 |
| fate_E10.S07.dE | 0.295 |
| fate_E10.S07.dO | 0.187 |
| fate_E10.S07.dD | 0.148 |
| fate_E10.S07.dS | 0.018 |
| fate_E10.S08.dT | 0.459 |
| fate_E10.S08.dE | 0.184 |
| fate_E10.S08.dO | 0.164 |
| fate_E10.S08.dD | 0.147 |
| fate_E10.S08.dS | 0.045 |

| quantity | value |
|---|---|
| fate_E10.S09.dT | 0.363 |
| fate_E10.S09.dE | 0.287 |
| fate_E10.S09.dO | 0.181 |
| fate_E10.S09.dD | 0.153 |
| fate_E10.S09.dS | 0.016 |
| fate_E10.S10.dT | 0.341 |
| fate_E10.S10.dE | 0.292 |
| fate_E10.S10.dO | 0.177 |
| fate_E10.S10.dD | 0.147 |
| fate_E10.S10.dS | 0.044 |
| fate_E10_all.dT | 0.377 |
| fate_E10_all.dE | 0.252 |
| fate_E10_all.dO | 0.186 |
| fate_E10_all.dD | 0.156 |
| fate_E10_all.dS | 0.029 |
| dose.S01.inf | 0.109 |
| dose.S01.dD | 4.425 |
| dose.S01.r2 | 0.949 |
| dose.S01.dose_maxT | 20 |
| dose.S01.dose_best | 2 |
| dose.S02.inf | 0.445 |
| dose.S02.dD | 3.234 |
| dose.S02.r2 | 0.955 |
| dose.S02.dose_maxT | 2 |
| dose.S02.dose_best | 2 |
| dose.S05.inf | 0.102 |
| dose.S05.dD | 5.21 |
| dose.S05.r2 | 0.658 |
| dose.S05.dose_maxT | 15 |
| dose.S05.dose_best | 15 |
| dose.S06.inf | 0.197 |
| dose.S06.dD | 3.887 |
| dose.S06.r2 | 0.931 |
| dose.S06.dose_maxT | 15 |
| dose.S06.dose_best | 2 |
| dose.S07.inf | 0.194 |
| dose.S07.dD | 4.106 |
| dose.S07.r2 | 0.929 |
| dose.S07.dose_maxT | 15 |
| dose.S07.dose_best | 2 |
| dose.S08.inf | 0.19 |
| dose.S08.dD | 3.85 |
| dose.S08.r2 | 0.945 |
| dose.S08.dose_maxT | 20 |
| dose.S08.dose_best | 8 |
| dose.S09.inf | 0.204 |
| dose.S09.dD | 4.388 |
| dose.S09.r2 | 0.926 |

| quantity | value |
|---|---|
| dose.S09.dose_maxT | 15 |
| dose.S09.dose_best | 2 |
| dose.S10.inf | 0.183 |
| dose.S10.dD | 3.394 |
| dose.S10.r2 | 0.923 |
| dose.S10.dose_maxT | 15 |
| dose.S10.dose_best | 2 |
| dose_predictors.n.rho_dD | 0.07143 |
| dose_predictors.n.rho_inf | 0.02381 |
| dose_predictors.taw.rho_dD | -0.2381 |
| dose_predictors.taw.rho_inf | 0.1905 |
| dose_predictors.se100.rho_dD | -0.1667 |
| dose_predictors.se100.rho_inf | 0.1905 |
| dose_predictors.dep300.rho_dD | 0.119 |
| dose_predictors.dep300.rho_inf | -0.04762 |
| dose_predictors.store.rho_dD | 0.1667 |
| dose_predictors.store.rho_inf | 0.1905 |
| dose_table.S01.dD | 4.425 |
| dose_table.S01.inf | 0.1094 |
| dose_table.S01.n | 1.353 |
| dose_table.S01.taw | 130 |
| dose_table.S01.se100 | 0.4445 |
| dose_table.S01.dep300 | 0.3794 |
| dose_table.S01.store | 14.8 |
| dose_table.S02.dD | 3.234 |
| dose_table.S02.inf | 0.4448 |
| dose_table.S02.n | 1.6 |
| dose_table.S02.taw | 105 |
| dose_table.S02.se100 | 0.3563 |
| dose_table.S02.dep300 | 0.4964 |
| dose_table.S02.store | 15.64 |
| dose_table.S05.dD | 5.21 |
| dose_table.S05.inf | 0.1017 |
| dose_table.S05.n | 1.89 |
| dose_table.S05.taw | 95 |
| dose_table.S05.se100 | 0.1647 |
| dose_table.S05.dep300 | 0.6274 |
| dose_table.S05.store | 17.88 |
| dose_table.S06.dD | 3.887 |
| dose_table.S06.inf | 0.1971 |
| dose_table.S06.n | 1.56 |
| dose_table.S06.taw | 150 |
| dose_table.S06.se100 | 0.4663 |
| dose_table.S06.dep300 | 0.4678 |
| dose_table.S06.store | 21.05 |
| dose_table.S07.dD | 4.106 |
| dose_table.S07.inf | 0.194 |
| dose_table.S07.n | 1.41 |

| quantity | value |
|---|---|
| dose_table.S07.taw | 220 |
| dose_table.S07.se100 | 0.6859 |
| dose_table.S07.dep300 | 0.3665 |
| dose_table.S07.store | 24.19 |
| dose_table.S08.dD | 3.85 |
| dose_table.S08.inf | 0.19 |
| dose_table.S08.n | 1.48 |
| dose_table.S08.taw | 160 |
| dose_table.S08.se100 | 0.417 |

# S8. Where every table and figure comes from

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Figure 1 | Schematic drawn by the script with numbers from the results | Manuscripts/reproducibility/analysis/p4_concept.py | Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig1_concept.pdf; numbers Manuscripts/results/p4_numbers.json |
| Table 1 | Table written by the script (tex/-data_table.tex) | Manuscripts/reproducibility/analysis/data_overview.py | Data/Comprehensive_Summary/00_summary_all_simulations |
| Figure 2 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/data_overview.py | Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/data/fig_data_overview.pdf |
| Figure 3 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p4_reach.py | Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig2_curves_reach.pdf; data Manuscripts/results/p4_tableA1_reach.csv |
| Table 2 | Table written by the script | Manuscripts/reproducibility/analysis/p4_reach.py | Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/results/p4_table2_laws.csv (sheet R_p4_table2_laws) |
| Figure 4 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p4_reach.py | Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig3_trigger_reach.pdf; data Manuscripts/results/p4_fig3_trigger_runs.csv, p4_tableA1_reach.csv |
| Figure 5 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p4_reach.py | Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig4_marginal_fate.pdf; data Manuscripts/results/p4_fig4a_fate_by_soil.csv, p4_fig4b_fate_by_method.csv |
| Figure 6 | Figure drawn by the script from the input data | Manuscripts/reproducibility/analysis/p4_reach.py | Data/Comprehensive_Summary/00_summary_all_simulations Manuscripts/figures/paper4/fig5_dose.pdf; data Manuscripts/results/p4_fig5_dose.csv |
| Figure 7 | Screen capture of SWAFL Desktop 1.1.0 (not regenerated) | - | Manuscripts/figures/shots_jpg/real_trigger_law.jpg; zaria_matric_potential.jpg; wb_step3_experiments.jpg |

| Item | How it is made | Script(s) | Input and output |
|---|---|---|---|
| Table 3 | Descriptive table (claims and the evidence cited in the text); no computation | - | |
| Table 4 | Table written by the script | Manuscripts/reproducibility/analysis/p4_reach.py | Data/Comprehensive_Summary/00_summary_all_simulations<br>Manuscripts/results/p4_tableA1_reach.csv (sheet R_p4_tableA1_reach) |
| Table 5 | Descriptive table (this map in short); no computation | - | |

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